Investigation of the Spatial and Temporal Variation of Temperature and Water Vapor in Lower Troposphere Using the Combined Backward Raman Scattering LiDAR and Lateral Scanning Raman Scattering LiDAR

Fan Yang, Fei Gao, Dengxin Hua, and Samo Stanič

***Abstract*— A novel Raman scattering LiDAR is proposed to achieve accurate measurements of atmospheric temperature and water vapor from the ground to the concerned height, in which a lateral scanning receiver is constructed to collect the lateral rotational and vibrational Raman scattering signals excited by the transmitted laser pulses from the backward Raman scattering LiDAR (BWRSL). The major advantages of the combined lateral scanning Raman scattering LiDAR (LSRSL) and BWRSL are the sharing of the transmitted laser and the measurements of temperature and water vapor without the effect of geometrical overlap function. To simplify the mechanical mounting and offer the commonality configuration, both polychromators in the LSRSL and BWRSL utilize the modular design with narrow-band interference filters to extract the high- and low-quantum-number transitions of the pure rotational Raman scattering signals and the vibrational Raman scattering signals of N_2 and H_2O. Different with the BWRSL in the profiling of Raman scattering signal in its dependence on the time elapsed after the transmission of the laser pulse, the profiling of the LSRSL is realized by the elevation angle scanning of the lateral receiver, in which four horizontally aligned telescopes mounted on a steerable frame are spatially separated to look at the vertical laser beam at a certain distance. The experimental measurements of both LSRSL system and BWRSL system are carried out with the temporal resolution of 16 min and the spatial resolution of 36 m. The retrieval results after data gluing shows that the combined BWRSL and LSRSL system can realize the detection of atmospheric temperature and water vapor from the ground to the height of 4.00 km. The error analysis shows that both deviation and statistical error of the atmospheric temperature are less than 1 K, and both deviation and statistical error of the water vapor mixing ratio are less than 0.5 g/kg within the detection height interval of 0–4.00 km.**

***Index Terms*— Atmospheric temperature, combined backward LiDAR and lateral scanning LiDAR, Raman scattering LiDAR, water vapor, without effect of overlap function.**

I. Introduction

ATMOSPHERIC temperature and water vapor are critical parameters for describing the physical state of the atmosphere. They directly influence the atmospheric heat balance and water vapor condensation processes, playing a crucial role in weather forecasting and environmental monitoring [1], [2], [3], [4]. Atmospheric temperature and water vapor within the lower troposphere closely influence human activities. High-precision measurements of the spatial and temporal variation of atmospheric temperature and water vapor from the ground to the concerned height not only help us study the atmospheric thermodynamics, and the global surface water cycle but also provide a valid scientific basis for the interrelationship between global warming and extreme weather [5], [6], [7], [8].

Raman LiDAR is a well-proven technique for detecting atmospheric temperature and water vapor with high spatial and temporal resolution, which depends on the extraction of the pure rotational Raman scattering spectra (PRRSS) or the vibrational Raman scattering spectra (VRSS) [9], [10], [11]. In temperature measurements, the transmission band in both high- and low-quantum-number rotational Raman channels need to be determined with great care because of the conflicting demands of the significant temperature dependence of the rotational Raman scattering signals, high signal intensity, and sufficiently high blocking of the elastic scattering signal in the rotational Raman channels [12], [13], [14], [15], [16], [17]. In the water vapor measurements, the final product is highly related to the ratio of the intensities of the VRSS of N_2 and H_2O molecules, not only because the signal intensities of the VRSS of N_2 and H_2O reflect the content of the corresponding molecules but also due to the

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relatively stable concentration distribution of N_2 molecules versus height in the atmosphere [18], [19], [20]. Due to the weak intensities of the PRRSS and the VRSS, researchers have focused not only on spectral discriminators for the extraction of the PRRSS and VRSS [21], [22], [23], [24] but also on the transmitter, receiver, and data acquisition to enhance the capability of detection range in the far-field [25], [26], [27], [28], [29].

The design of the traditional Raman scattering LiDAR uses the monostatic structure, in which the receiver is configured to collect the Raman backscattering signals in its dependence on the time elapsed after the transmission of the laser pulses. Whatever the noncoaxial or coaxial structure between the transmitter and the receiver, the geometrical overlap function with the value less than 1 cannot be avoid in its design, which limits the accurate measurement in the near-ground atmosphere [30]. The lateral scattering LiDAR in the aerosol measurement gives an idea to overcome the effects of the geometrical overlap function of the backward LiDAR [31], [32], [33], [34], [35], [36]. However, due to the intensities of the PRRSS and VRSS with three magnitudes weaker than the Mie–Rayleigh scattering signals, it is really difficult to record the lateral scattering signals of the PRRSS or the VRSS using charge-coupled device (CCD) for temperature and water vapor measurements.

In this article, we report a combined backward Raman scattering LiDAR (BWRSL) and lateral scanning Raman scattering LiDAR (LSRSL) for simultaneous measurements of atmospheric temperature and water vapor from the ground to the concerned height. The major advantages of the combined BWRSL and LSRSL system are the sharing of the transmitted laser and the measurement without the effect of geometrical overlap function. The working principle, system setup, data processing, and preliminary measurements are presented successively in Sections II–V, respectively.

II. Theory

A. Working Principle

The BWRSL system utilizes a monostatic structure between the transmitter and the receiver, whose working principle is shown in Fig. 1(a). The pulsed laser transmits the laser pulse vertically to the atmosphere and interacts with the particles and molecules, while the receiver is configured to collect the inelastic Raman backscattering signals based on the time elapsed after the transmission of the laser pulse. However, the no-overlap region, with no overlap and the partial-overlap region with partial overlap between the transmitting field and the receiving filed, called the geometrical overlap function, limits the detection of atmospheric information in the near-ground atmosphere. The geometrical overlap function of the BWRSL system varies from several 10 m to several 100 m depending on the system parameters, including the divergence angle of laser beam, the field of view (FOV) of telescope receiver, and the optical axis separation between the transmitter and the receiver, and so on. It is obvious that within the no-overlap region and partial-overlap region, the backward telescope cannot pick up the total Raman backscattering signals for temperature and water vapor retrieval.

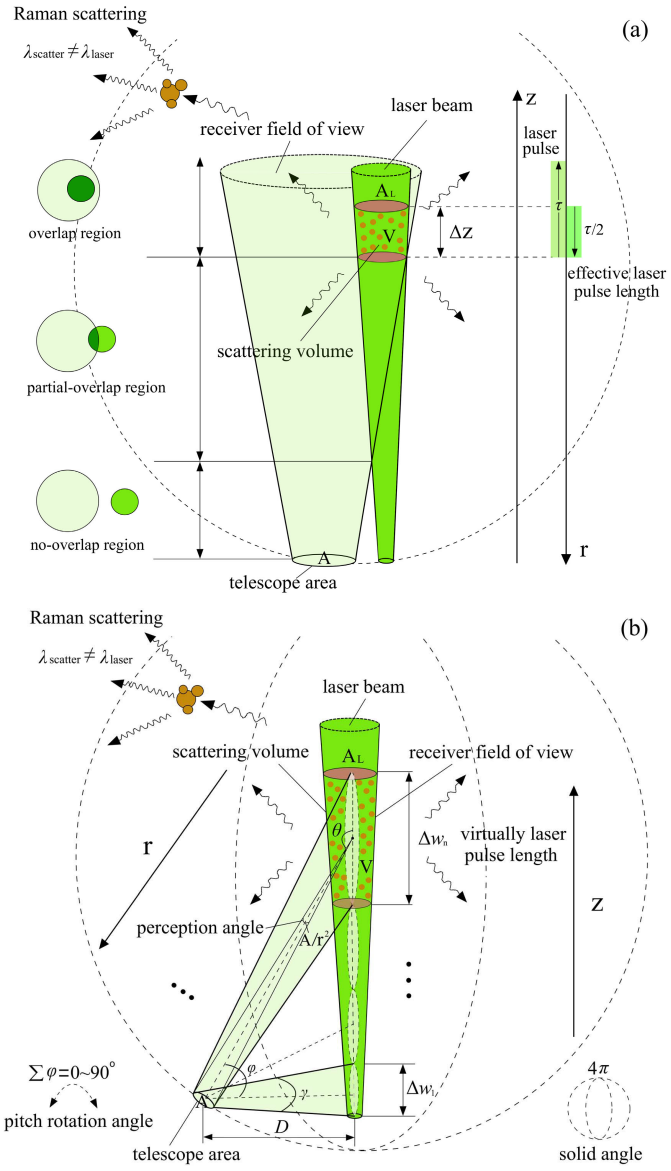


Fig. 1. Working principle of the Raman scattering LiDAR. (a) Technique of BWRSL. (b) Technique of LSRSL.

The LSRSL system adopts a bistatic structure between the transmitter and the receiver, whose working principle is shown in Fig. 1(b). A lateral receiving system is spatially separated to look at the vertical-transmitted laser beam at a certain distance, which is used to collect the lateral inelastic Raman scattering signals. The profiling of lidar returns in the LSRSL system is performed by the elevation angle scanning of the lateral receiving system, resulting in the accurate measurement of atmospheric temperature and water vapor from the ground to the concerned height and to overcome the effect of the geometrical overlap function of the BWRSL system. Considering the working principle of the LSRSL technique, both pulsed laser and continuous laser can be taken as the transmitter.

To realize the investigation of the spatial and temporal variation of temperature and water vapor in the lower troposphere without the effect of no-overlap region and partial-overlap region, a combined Raman lidar system is presented

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using both BWRSL technique and LSRSL technique, whose major advantages are the sharing of the transmitted laser and the complementary of atmospheric temperature and water vapor in the measurement height from the ground to the concerned height.

B. Raman LiDAR Equation

The LiDAR equations for both BWRSL system and LSRSL system can be expressed as

$$E_X\left(z, \lambda_0, \lambda_X^{\mathrm{Ram}}, \theta\right) = K \frac{O(z)}{z \cdot \frac{-z}{\cos\theta}} \beta\left(\lambda_0, \lambda_X^{\mathrm{Ram}}, z, \theta\right) T_z\left(z, \lambda_0\right) T_X\left(z, \lambda_X^{\mathrm{Ram}}, \theta\right) \tag{1}$$

where $E_X(z, \lambda_0, \lambda_X^{\mathrm{Ram}}, \theta)$ is the received power of the Raman scattering signal at the height z with the scattering angle θ, K is the system constant, $O(z)$ is the geometric overlap function, $\beta(\lambda_0, \lambda_X^{\mathrm{Ram}}, z, \theta)$ is the scattering coefficient at height z with the scattering angle θ, $T_z(z, \lambda_0)$ is the outgoing atmospheric transmittance, $T_X(z, \lambda_X^{\mathrm{Ram}}, \theta)$ is the returning atmospheric transmittance at the Raman spectral wavelength λ_X^{Ram} with the scattering angle θ. X denotes the Raman spectrum. BWRSL system can be taken as a special type of LSRSL system with the scattering angle θ of 180°. Moreover, for the LSRSL system with the scattering angle θ between 90° and 180°, $O(z)$ equals 1.

Not only the items with the scattering angle θ are different but also the item of the system constant K is different for the BWRSL equation and the LSRSL equation. Considering the system configuration and system parameters, the system constant K can be expressed as

$$\begin{cases} K = E_0 A \eta \dfrac{c\tau}{2}, & \text{for BWRSL} \\ K = E_0 A \eta \Delta w, & \text{for LSRSL} \end{cases} \tag{2}$$

where E_0 is the emitted laser power, A is the effective receiving area, η is the overall system efficiency, c is the light speed, τ is the pulsewidth. $c\tau/2$ is called the pulselength in the BWRSL system, expressing the length of the volume light illuminated by the laser pulse at a fixed time. Similarly, Δw is called the virtual pulselength in the LSRSL system, expressing the length of the volume light received instantaneously within the FOV of the lateral telescope and varies with the horizontal distance between transmitter and receiver D, the FOV of telescope γ, and scattering angle θ as

$$\Delta w = \frac{D}{\sin^2\theta}\gamma. \tag{3}$$

The Raman scattering coefficient $\beta(\lambda_0, \lambda_X^{\mathrm{Ram}}, z, \theta)$ can be expressed as

$$\beta\left(\lambda_0, \lambda_X^{\mathrm{Ram}}, z, \theta\right) = N(z)\sigma_X(\pi) f_X(\theta) \tag{4}$$

where $N(z)$ is atmospheric molecular density, $\sigma_X(\pi)$ is the differential backscatter cross section for the Raman spectrum X. $f_X(\theta)$ is the normalized phase function of molecular scattering at the scattering angle θ for the Raman spectrum X. When the scattering angle θ is 180°, $f_X(\theta)$ takes the value of 1 for Raman backscattering spectra in the BWRSL system.

For both BWRSL technique and LSRSL technique, the outgoing atmospheric transmittance $T_z(z, \lambda_0)$ expresses the light attenuation vertically along the transmitting path from the ground to the height z at the transmitter wavelength λ_0, which is expressed as

$$T_z(z, \lambda_0) = \exp\left\{ -\int_0^z \left[\alpha_{\mathrm{aer}}(z, \lambda_0) + \alpha_{\mathrm{mol}}(z, \lambda_0)\right] \mathrm{d}z \right\} \tag{5}$$

where $\alpha(z, \lambda_0)$ is the extinction coefficient including both aerosol and molecular contributions. Although the returning atmospheric transmittance $T_X(z, \lambda_X^{\mathrm{Ram}}, \theta)$ in both BWRSL system and LSRSL system is a function of extinction coefficient at the Raman spectral wavelength λ_X^{Ram}, the expression equation is different due to the different returning light path and expressed as

$$\begin{cases} T_X(z, \lambda_X^{\mathrm{Ram}}) = \exp\left\{ -\int_0^z \left[\alpha_{\mathrm{aer}}(z, \lambda_X^{\mathrm{Ram}}) + \alpha_{\mathrm{mol}}(z, \lambda_X^{\mathrm{Ram}})\right] \mathrm{d}z \right\}, & \text{for BWRSL} \\ T_X(z, \lambda_X^{\mathrm{Ram}}, \theta) = \exp\left\{ \dfrac{1}{\cos\theta} \displaystyle\int_0^z \left[\alpha_{\mathrm{aer}}\left(z, \lambda_X^{\mathrm{Ram}}\right) + \alpha_{\mathrm{mol}}\left(z, \lambda_X^{\mathrm{Ram}}\right)\right] \mathrm{d}z \right\}, & \text{for LSRSL.} \end{cases} \tag{6}$$

The final LiDAR equation for expressing the BWRSL technique can be expressed as

$$E_X\left(z, \lambda_0, \lambda_X^{\mathrm{Ram}}\right) = E_0 \frac{c\tau}{2} A K \frac{O(z)}{z^2} N(z)\sigma_X(\pi) \times \exp\left\{ \begin{array}{l} -\displaystyle\int_0^z \left[\alpha_{\mathrm{aer}}(z, \lambda_0) + \alpha_{\mathrm{mol}}(z, \lambda_0)\right] \mathrm{d}z \\ -\displaystyle\int_0^z \left[\alpha_{\mathrm{aer}}\left(z, \lambda_X^{\mathrm{Ram}}\right) + \alpha_{\mathrm{mol}}\left(z, \lambda_X^{\mathrm{Ram}}\right)\right] \mathrm{d}z \end{array} \right\} \tag{7}$$

where $O(z)$ in (7) represents the geometrical overlap function and achieves 1 above the height of 0.87 km in our designed BWRSL system presented in Section III-A. The final LiDAR equation for expressing the LSRSL technique can be expressed as

$$E_X\left(z, \lambda_0, \lambda_X^{\mathrm{Ram}}, \theta\right) = E_0 A K \frac{\gamma}{D} N(z)\sigma_X(\pi) f_X(\theta) \times \exp\left\{ \begin{array}{l} -\displaystyle\int_0^z \left[\alpha_{\mathrm{aer}}(z, \lambda_0) + \alpha_{\mathrm{mol}}(z, \lambda_0)\right] \mathrm{d}z \\ +\dfrac{1}{\cos\theta} \displaystyle\int_0^z \left[\alpha_{\mathrm{aer}}\left(z, \lambda_X^{\mathrm{Ram}}\right) + \alpha_{\mathrm{mol}}\left(z, \lambda_X^{\mathrm{Ram}}\right)\right] \mathrm{d}z \end{array} \right\}. \tag{8}$$

C. Raman Scattering Spectra and Their Extraction Methods

In temperature measurement using the pure rotational Raman LiDAR technique, two detection channels are defined to retrieve the atmospheric temperature, which is called the

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high-quantum-number rotational Raman channel to extract the high-quantum-number transitions of PRRSS and the low-quantum-number rotational Raman channel to extract the low-quantum-number transitions of PRRSS, respectively. The signal intensities of the high- and low-quantum-number transitions of the PRRSS present different variation tendencies (decrease and increase) with the increment of temperature, as shown in Fig. 2(a). Temperature sensitivity $\Delta\sigma(J)$ is introduced for further analysis of the dependence between the high- and low-quantum-number transitions of the PRRSS and atmospheric temperature, which is expressed as

$$\Delta\sigma(J) = \frac{\sigma(J, T_1) - \sigma(J, T_2)}{T_1 - T_2} \tag{9}$$

where $\sigma(J, T_1)$ and $\sigma(J, T_2)$ are the differential backscatter cross section with the spectral line J at atmospheric temperature $T_1 = 200$ K and $T_2 = 310$ K, respectively. The temperature sensitivity of PRRSS with the excitation wavelength of 532.10 nm is shown in Fig. 2(b). The temperature sensitivity of high-quantum-number transitions of PRRSS is positively correlated with temperature, while the temperature sensitivity of low-quantum-number transitions of PRRSS is negatively correlated with temperature in both Stokes branch and anti-Stokes branch. To enhance the retrieval accuracy of temperature, the spectral lines with the most significant temperature sensitivity varied in both Stokes branch and anti-Stokes branch (positive and negative correlations) should be selected and extracted by the polychromator of the Raman LiDAR. Meanwhile, the spectral lines in the low-quantum-number rotational Raman channel should be selected as far as possible from the transmitter wavelength to minimize the interference of elastic scattering signals. Therefore, in the anti-Stokes branch, $J_{\text{high}} = 13$ is chosen as the central spectral line of high-quantum-number transitions of PRRSS, and $J_{\text{low}} = 5$ is chosen as the central spectral line of low-quantum-number transitions of PRRSS. While in the Stokes branch, $J_{\text{high}} = 15$ is chosen as the central spectral line of high-quantum-number transitions of PRRSS, and $J_{\text{low}} = 5$ is chosen as the central spectral line of low-quantum-number transitions of PRRSS.

To further explore the advantages and disadvantages of the selected spectral lines of the high- and low-quantum-number transitions in the Stokes branch and the anti-Stokes branch of PRRSS, the statistical temperature uncertainties $\Delta T_{1-\sigma}$ of the high- and low-quantum-number transitions of PRRSS are calculated considering both N_2 and O_2 contributions. The statistical temperature uncertainty $\Delta T_{1-\sigma}$ can be expressed as

$$\Delta T_{1-\sigma} = \frac{\partial T}{\partial Q}\Delta Q \approx \frac{(T_1 - T_2)}{(Q_1 - Q_2)} Q \sqrt{\frac{1}{E_{\text{low}}} + \frac{1}{E_{\text{high}}}} \tag{10}$$

where $Q = E_{\text{high}}/E_{\text{low}}$ is the ratio of signal intensities of the high-quantum-number rotational Raman channel and low-quantum-number rotational Raman channel. Fig. 3 shows the calculation results of the statistical temperature uncertainty with the spectral lines of J_{high} and J_{low} in both Stokes branch and anti-Stokes branch for different temperature pairs of (260, 265 K), (280, 285 K), and (300, 305 K), which corresponds to

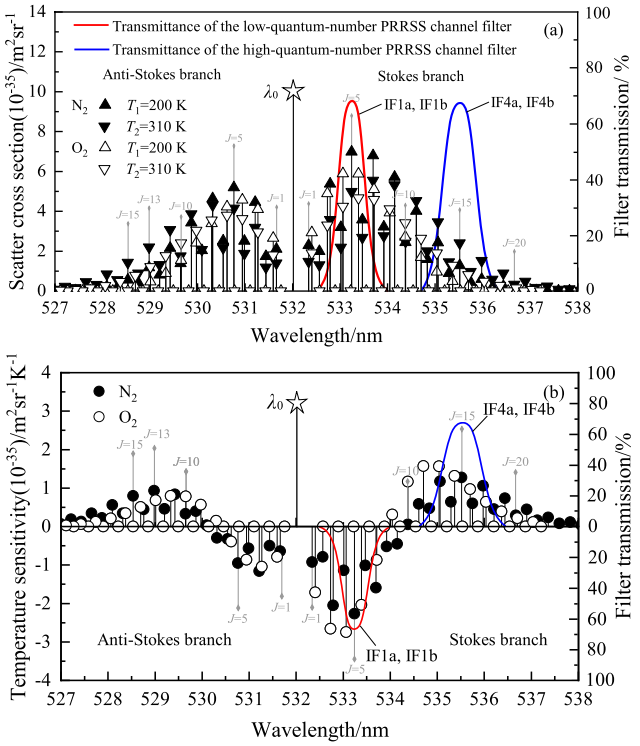


Fig. 2. PRRSS spectral lines of N_2 and O_2 with the excitation wavelength of 532.10 nm and the selected filter transmission (IF1a, IF1b and IF4a, IF4b with CWL of 533.24 and 535.52 nm) used in the polychromator of the rotational Raman LiDAR. (a) PRRSS of N_2 and O_2. (b) Temperature sensitivity of PRRSS of N_2 and O_2.

the heights of 4.5, 1.5, and 0 km from the standard atmospheric model.

It can be seen from Fig. 3, that the minimum statistical temperature uncertainty (□, $\Delta T_{1-\sigma}$ =1.0 K) increases gradually with the increased temperature in the high-quantum-number transitions of PRRSS in both anti-Stokes branch and Stokes branch, while it does not change in the low-quantum-number transitions of PRRSS. By selecting the spectral lines in the high-quantum-number transitions of PRRSS with larger spectral interval relative to the transmitter wavelength, the temperature sensitivity of the LiDAR system can be improved. The spectral lines in high-quantum-number transitions of PRRSS are with more temperature sensitivity and weaker signal intensity, while the spectral lines in low-quantum-number transitions of PRRSS are with stronger signal intensities and smaller spectral interval relative to the elastic scattering channel.

Considering that the signal intensity, temperature sensitivity, and statistical temperature uncertainty of the corresponding spectral lines in the Stokes branch are better than those in the anti-Stokes branch, the spectral lines in the Stokes branch of PRRSS are selected to construct the detection channels in the rotational Raman LiDAR. Moreover, to increase the signal-to-noise ratio, the high-quantum-number rotational Raman channel is ultimately determined to consist of five N_2 spectral lines of $J_{\text{high}} = 13, 14, 15, 16, 17$, and three O_2 spectral lines of $J_{\text{high}} = 15, 17, 19$ in the Stokes branch with the central wavelength (CWL) of 535.52 nm and full width at

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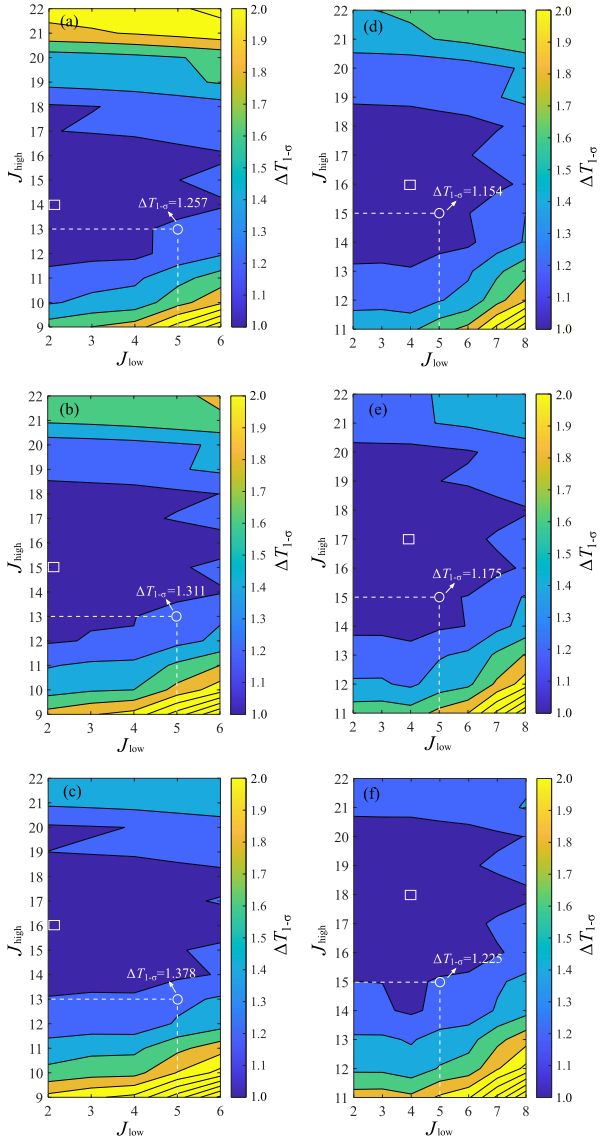


Fig. 3. Calculation of the statistical temperature uncertainties $\Delta T_{1-\sigma}$ with the variation of J_{high} and J_{low} at different temperature pairs. (a) Anti-Stokes branch with temperature pair of (260, 265 K). (b) Anti-Stokes branch with temperature pair of (280, 285 K). (c) Anti-Stokes branch with temperature pair of (300, 305 K). (d) Stokes branch with temperature pair of (260, 265 K). (e) Stokes branch with temperature pair of (280, 285 K). (f) Stokes branch with temperature pair of (300, 305 K). The symbol of □ marks the scaled minimum values of $\Delta T_{1-\sigma}$ and the circle (◯) marks our selection of the PRRSS.

half maximum (FWHM) of 0.87 nm, and the low-quantum-number rotational Raman channel consists of three N_2 spectral lines of $J_{low} = 4, 5, 6$ and two O_2 spectral lines $J_{low} = 5, 7$ in the Stokes branch with the CWL of 533.24 nm and FWHM of 0.45 nm. As both high- and low-quantum-number rotational Raman channels are susceptible to interference from the elastic scattering signals, the interference filters with the out of band rejection rate more than six are selected separately to obtain a higher optical density and reduce the interference from elastic scattering.

In water vapor measurement using the vibrational Raman LiDAR technique, the VRSS of N_2 and H_2O molecules are selected to detect the water vapor mixing ratio, considering

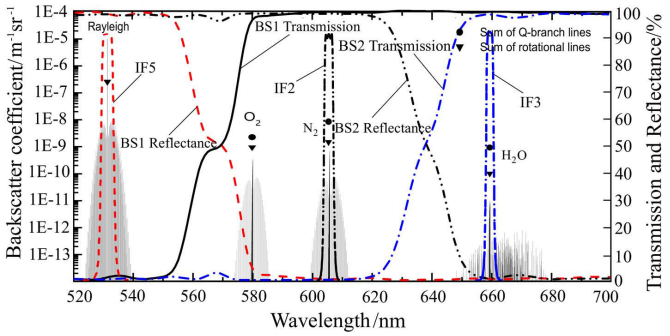


Fig. 4. Transmission and reflectance curves of the interference filters (IF2, IF3, IF5 corresponding to CWL of 607.44, 660.43, and 532.10 nm) and dichroic beam splitter (BS1, BS2) used in the polychromator of the vibrational Raman LiDAR, together with the vibrational Raman spectrum excited by the transmitter wavelength of 532.10 nm.

that the signal intensities of N_2 and H_2O reflect the content of the corresponding molecules and the relatively stable concentration distribution of N_2 molecules versus height in the atmosphere. The lateral VRSS of N_2 and H_2O molecules have the same frequency shift of 2331 and 3652 cm^{-1} with the backward VRSS. With the transmitter wavelength of 532.10 nm, the CWL of the VRSS of N_2 and H_2O molecules are 607.44 and 660.43 nm, respectively. Fig. 4 shows the VRSS with the transmitter wavelength of 532.10 nm and the transmission and reflectance curves of the narrowband interference filters (IF2, IF3) and dichroic beam splitter (BS1, BS2) used in the polychromator of the vibrational Raman LiDAR. The narrowband interference filter (IF5) is used to extract the elastic scattering spectrum with the CWL of 532.10 nm and FWHM of 1.80 nm.

III. LiDAR System

A. Instrument Description

Fig. 5 presents a schematic of the combined Raman LiDAR system designed for measuring atmospheric temperature and water vapor, incorporating both BWRSL and LSRSL systems. The high-power Nd: YAG pulsed laser is utilized as the common transmitter for both systems, which emit a radiation light of the pulse energy of 200 mJ at the wavelength of 532.10 nm with a repetition rate of 10 Hz and the pulsewidth of 10 ns. A home-made 5× beam expander is equipped to reduce the divergence angle of the output laser beam to be <0.1 mrad. A 2-D reflector mirror (RM1) is used to guide the output laser beam to the atmosphere vertically, where it interacts with atmospheric particles and molecules and scattered elastically and inelastically.

For the BWRSL system with a vertical-positioned Cassegrain telescope (Telescope5), a monostatic noncoaxial structure between the transmitter and the receiver is adopted, which results in a distance of 235 mm between their axes. The Cassegrain telescope has a diameter of 203 mm and a focal length of 2000 mm. The collected light is collimated by a collimating lens (CL5) and then redirected by a reflecting mirror (RM2) into the polychromator of the BWRSL system, which is designed by the beam splitter and the narrow-band

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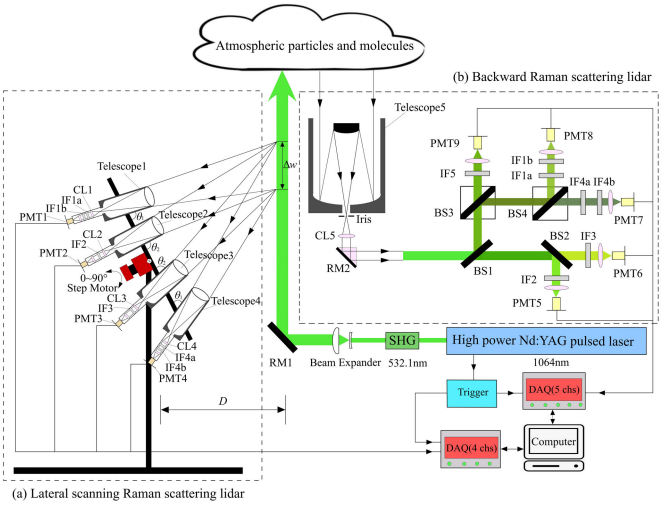


Fig. 5. Schematic of the combined BWRSL and LSRSL system. (a) Receiver optics in LSRSL system. (b) Receiver optics in BWRSL system.

interferometric filters for separation and extraction of the backward scattering signals of PRRSS and VRSS.

For the LSRSL system with a lateral scanning receiver, four Kepler telescopes with the same specifications of the diameter of 90 mm and the focal length of 900 mm are spatially separated to look at the laser beam from a certain distance D of 70 m, to collect the lateral scattering signals of PRRSS and VRSS from the ground to the height of 1.37 km. In order to make the lateral receiver system with the elevation angle scanning capability, those four telescopes are put on a horizontal common frame to make them with the same elevation angle, whose elevation angle is adjustable within the angular interval of 0°–90° using a step motor with 0.1° angular resolution. Such configuration setup produces a bistatic LiDAR configuration and does not suffer from the overlap effect as in the BWRSL system any more. To make those four telescopes look at the same part of the laser beam simultaneously, Telescope1 and Telescope4 are set to have the titled angle θ_1 of 89.75° and the separation distance of 600 mm, and Telescope2 and Telescope3 are set to have the titled angle θ_2 of 89.92° and the separation distance of 200 mm, as shown in Fig. 5. By the end of telescope, the polychromator system with the corresponding narrow-band interference filter is designed to extract the lateral scattering signal of PRRSS and VRSS. The profiling of LiDAR returns in the LSRSL system is performed by the elevation angle scanning of the lateral receiving system, in which the intensities of the lateral Raman scattering signals at each setting of elevation angles are sampled and analyzed.

Due to their working principles, the BWRSL system and the LSRSL system have the complementary detection height, where the BWRSL system can realize the atmospheric detection at higher heights, and the LSRSL system can realize the atmospheric detection at lower heights. Therefore, the combined BWRSL and LSRSL system can result in the detection height of atmospheric temperature and water vapor from the ground to the concerned height.

B. Polychromator Design and Adjustment

Although the working principles of the BWRSL technique and LSRSL technique are different, the spectra of the high- and low-quantum-number transitions of PRRSS for temperature measurement and the VRSSS of N_2 and H_2O for water vapor measurement are the same. Therefore, the modular design is adopted to improve the reliability and replaceability of the polychromator, which uses the narrow-band interference filters as the key component in the spectral extraction. The IF1a and IF1b have a peak transmittance of >60% at the CWL of 533.24 nm and a FWHM of 0.45 nm to extract the signals of the low-quantum-number transitions of the PRRSS, and provide a rejection ratio of >8 orders of magnitude to signals out of band. The IF4a and IF4b have a peak transmittance of >60% at the CWL of 535.52 nm and a FWHM of 0.87 nm to extract the signals of the high-quantum-number transitions of the PRRSS, and provide a rejection ratio of >8 orders of magnitude to signals out of band. The IF2 has a peak transmittance of >90% at the CWL of 607.44 nm and a FWHM of 1.80 nm to extract the signals of the VRSS of N_2, and provides a rejection ratio of >6 orders of magnitude to signals out of band. The IF3 has a peak transmittance of >90% at the CWL of 660.43 nm and a FWHM of 1.80 nm to extract the signals of the VRSS of H_2O, and provides a rejection ratio of >6 orders of magnitude to signals out of band. Both BWRSL system and the LSRSL system convert the filtered light signals to electrical signals by the photomultiplier tube (PMT) with a quantum efficiency of ∼25% at 533.24 nm, ∼25% at 535.52 nm, ∼22% at 607.44 nm, and ∼20% at 660.43 nm.

Fig. 6 shows a structural diagram of the polychromator setup for the LSRSL system. The polychromator of the LSRSL system is integrated with the tail of the Kepler telescope, which is assembled sequentially by the iris, collimating lens, narrow-band interference filters (IF1a and IF1b, IF2, IF3, IF4a, and IF4b), and focusing lens. All the optical components are installed in an optical cage-type light-blocking cylinder with a clamping ring. The connection between the polychromator and the telescope is secured using SM1 internal and external threads. The PMT is put on the focus of the focusing lens and connected with the light-blocking cylinder using a home-made auxiliary mechanical device. The output of the PMT is amplified and converted the current signal to the voltage signal using an operational amplification circuit for the initial signal processing. The length of the compact polychromator of the LSRSL system is limited to 210 mm. The iris aperture is 3.6 mm, which sets the FOV of the telescope receiver to 4 mrad. The position of the collimating lens can be adjusted within the optical cage-type light-blocking cylinder to ensures the parallel light passing through the narrow-band interference filter.

In the BWRSL system, only one Cassegrain telescope is utilized as the receiving optics to collect the Raman backscattering signals. Therefore, the beam splitter is selected to construct four channels in the polychromator of the BWRSL system, which uses the narrow-band interference filter as a key component for spectral extraction as well. The optical cage-type light-blocking cylinder designed in the LSRSL

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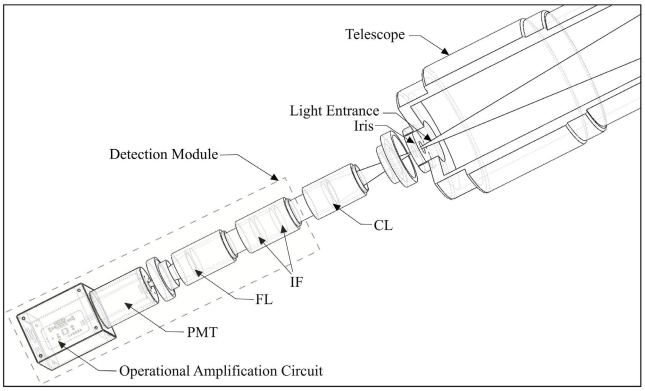


Fig. 6. Structural diagram of the polychromator of the LSRSL system. CL: collimating lens; IF: narrow-band interference filter; FL: focusing lens; PMT: photomultiplier tube.

system is utilized to integrate the optics in the polychromator of the BWRSL as well, including the narrow-band interference filter, focusing lens and PMT for extracting and detecting the desired Raman scattering signals.

Fig. 7(a) shows a top view of the polychromator setup for the BWRSL system. The backscattering light is collected by the Cassegrain telescope (Telescope5) with a diameter of 203 mm and a focal length of 2000 mm, and then passing through an iris with a diameter of 1 mm, which limits the FOV of receiving optics to 0.5 mrad. The collimating lens (CL5) is taken to make the collected LiDAR returns be collimated light into the polychromator via the light inlet. A reflector mirror (RM2) is used to change the light direction with 90° to make the polychromator smaller. The dichroic beam splitter BS1 separates the RM2-reflected light into two parts: the reflected part is used to construct high- and low-quantum-number rotational Raman channels for temperature measurement, while the transmitted part is used to construct the detection channels of VRSS of N_2 and H_2O for water vapor measurement. The 10% of the reflected light by the BS1 is transmitted by the cubic beam splitter BS3 to construct the Mie channel for checking the system detection performance, while the other 90% of the reflected light by the BS1 is reflected by the cubic beam splitter BS3 and transmitted to the cubic beam splitter BS4 with half reflection and half transmittance to construct the high- and low-quantum-number rotational Raman channels. The transmitted light by the BS1 is transmitted to the dichroic beam splitter BS2 to construct the detection channels of VRSS of N_2 and H_2O. The exits (EX5, EX6, EX7, EX8, and EX9) of each detection channel are connected with the detection module for light spectral extraction and detection, which realizes the extraction of the desired Raman spectrum by the selection of the interference filters. The beam splitters are put in the optical box, which are connected to each other by the optical connecting cylinder. The detailed information of key components in the combined BWRSL and LSRSL system is shown in Table I.

TABLE I
KEY PARAMETERS SUMMARIZED FOR THE COMBINED BWRSL AND LSRSL SYSTEM

Transmitter	
Laser	Nd: YAG, 10 Hz, 200 mJ/Pulse@532.10 nm;
BE	5x; beam divergence:<0.1 mrad;
Receiver	
Lateral	d=90 mm, f=900 mm, FOV=4 mrad;
Backward	d=203 mm. f=2000 mm, FOV=0.5 mrad;
Spectroscopic	
Filters	IF1: CWL=533.24 nm, FWHM=0.45 nm, Tp>60%, outband OD>4; IF2: CWL=607.44 nm, FWHM=1.80 nm, Tp>90%, outband OD>6; IF3: CWL=660.43 nm, FWHM=1.80 nm, Tp>90%, outband OD>6; IF4: CWL=535.52 nm, FWHM=0.87 nm, Tp>60%, outband OD>4; IF5: CWL=532.10 nm, FWHM=1.80 nm, Tp>90%, outband OD>6;
Beam Splitters	BS1: TR>95%@584nm-800 nm, RE>95%@380nm-550 nm; BS2: TR>95%@655nm-700 nm, RE>95%@580nm-621 nm; BS3: cubic, RE:TR=9:1; TR, RE>95%; BS4: cubic, RE:TR=1:1; TR, RE>95%;
Pitch rotation platform	
Step Motor	Spinning speed: 0.02~8 s, Positioning accuracy: ±0.1°, Elevation angle/: 0 ~90°;
Data Acquisition	
Tektronix MSO46	Sampling Rate: 125 MS/s, Record length: 0.01 M;
Detectors	
Hamamatsu PMT H11901	Quantum Efficiency: ~25%@533.24 nm, ~22%@607.44 nm, ~20%@660.43 nm, ~25%@535.52 nm; Gain: 2×10^6;

BE: beam expander; Tp: peak transmittance; OD: optical density; TR: transmittance; RE: reflectivity;

IV. DATA PROCESSING

A. Inversion Algorithms of Atmospheric Temperature and Water Vapor

In fact, the BWRSL system can be taken as the special type of the LSRSL system with the scattering angle of 180°. Thus, we take both BWRSL technique and LSRSL technique together in the expression of inversion algorithm of atmospheric temperature and water vapor. Atmospheric temperature $T(z)$ is retrieved by the fit quadratic function $Q(T, z)$, which is the ratio of signals of the high- and low-quantum-number transitions of the PRRSS and is expressed as

$$Q(T,z) = \exp\left[A_T + \frac{B_T}{T(z)} + \frac{C_T}{T^2(z)}\right]\Delta f_{\mathrm{R-Ram}}(\theta)\Delta T_X \times \left(z, \lambda_{\mathrm{low}}^{\mathrm{R-Ram}}, \lambda_{\mathrm{high}}^{\mathrm{R-Ram}}, \theta\right) \quad (11)$$

where A_T, B_T, and C_T are the calibration coefficients in the quadratic fitting, $\Delta f_{\mathrm{R-Ram}}(\theta)$ is the correction function of phase function of molecular scattering with scattering angle θ, $\Delta T_X(z, \lambda_{\mathrm{low}}^{\mathrm{R-Ram}}, \lambda_{\mathrm{high}}^{\mathrm{R-Ram}}, \theta)$ is the correction function of atmospheric transmittance along the return path with the

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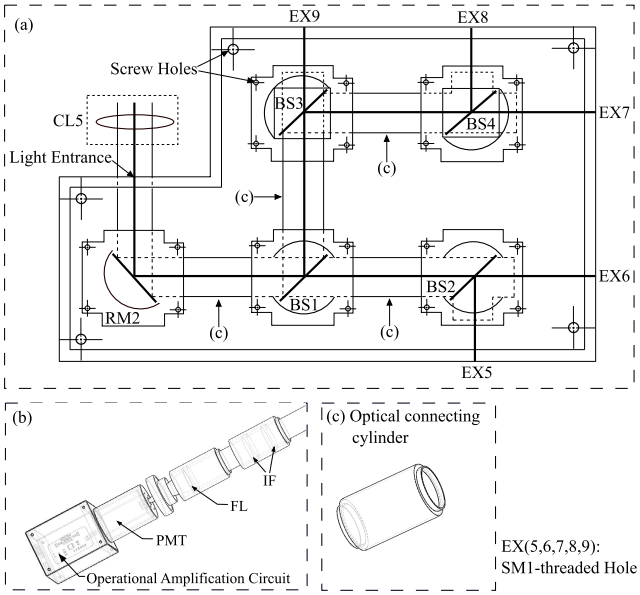


Fig. 7. Structural diagram of polychromator for the BWRSL system. (a) Top view of the polychromator for the BWRSL system receiver. (b) Schematic of detection module. (c) Cage-type connecting cylinder. RM: reflector mirror; BS: beam splitter.

scattering angle θ due to the wavelength difference between the high- and low-quantum-number transitions of the PRRSS, which is negligible as the wavelengths of the high- and low-quantum-number transitions of the PRRSS are so close in both BWRSL system and LSRSL system.

Atmospheric water vapor mixing ratio $W(z)$ is retrieved by the ratio of the signals of the VRSS of N_2 and H_2O and is expressed as

$$W(z) = C_W \frac{E_{H_2O}\left(z, \lambda_0, \lambda_{H_2O}^{\text{V-Ram}}, \theta\right)}{E_{N_2}\left(z, \lambda_0, \lambda_{N_2}^{\text{V-Ram}}, \theta\right)} \Delta f_{\text{V-Ram}}(\theta) \Delta T_X \times \left(z, \lambda_{H_2O}^{\text{V-Ram}}, \lambda_{N_2}^{\text{V-Ram}}, \theta\right) \tag{12}$$

where C_W is calibration coefficients for water vapor mixing ratio retrieval, $\Delta f_{\text{V-Ram}}(\theta)$ is the correction function of phase function of molecular scattering with scattering angle θ, $\Delta T_X(z, \lambda_{H_2O}^{\text{V-Ram}}, \lambda_{N_2}^{\text{V-Ram}}, \theta)$ is the correction function of atmospheric transmittance along the return path with scattering angle of θ due to the wavelength difference between the VRSS of N_2 and H_2O molecules. As the wavelength difference between the VRSS of N_2 and H_2O is more than 50 nm, the correction function ΔT_X in (12) needs to be corrected, which is done using the retrieved aerosol extinction coefficient from the real-time detection of the vibrational Raman scattering signal of N_2 in the combined BWRSL and LSRSL system. The detailed correction procedure will be presented in another article, which is out of the scope of this study. In the LSRSL system, the scattering angle θ in all the four telescopes has the same setting, resulting in the correction coefficients $\Delta f_{\text{R-Ram}}(\theta)$ in (11) and $\Delta f_{\text{V-Ram}}(\theta)$ in (12) equal to one. In BWRSL system, the scattering angle θ can be recognized as the special value of 180°, resulting in a nonexist correction coefficient and is defined as a constant value of one.

B. Analysis of Statistical Error of Atmospheric Temperature and Water Vapor

In atmospheric temperature measurement, temperature measurement sensitivity Θ is defined as the fractional change in the response function with a 1 K temperature change for a given temperature condition [37]

$$\Theta = \frac{1}{Q(T,z)} \frac{\partial Q(T,z)}{\partial T}. \tag{13}$$

The statistical error in temperature determined by the temperature measurement sensitivity and signal-to-noise ratio of the high- and low-quantum-number rotational Raman channels can be expressed as

$$\Delta T = \left[\left(\frac{1}{\text{SNR}_{\text{high}}} \right)^2 + \left(\frac{1}{\text{SNR}_{\text{low}}} \right)^2 \right]^{1/2} \cdot \frac{1}{\Theta} \tag{14}$$

where SNR_{high} and SNR_{low} are the signal-to-noise ratio of high- and low-quantum-number rotational Raman channels, respectively.

In water vapor measurement, the statistical error of the water vapor mixing ratio is mainly generated by the signal-to-noise ratio of the detection channels of the VRSS of N_2 and H_2O, which can be expressed as

$$\Delta W = \left[\left(\frac{1}{\text{SNR}_{N_2}} \right)^2 + \left(\frac{1}{\text{SNR}_{H_2O}} \right)^2 \right]^{1/2} \tag{15}$$

where SNR_{N2} and SNR_{H2O} are the signal-to-noise ratio of the detection channels of VRSS of N_2 and H_2O, respectively.

In LiDAR detection, the shot noise is the main noise source, as the PMT is employed as the photodetector. Moreover, the background noise includes the dark current of the photodetector, the sky stray light and the various noises generated by the load resistance as well. Therefore, the signal-to-noise ratio in each detection channel using analog detection mode can be expressed as

$$(\text{SNR})_X = \frac{\sqrt{n} \cdot i_s^2}{(i_s + i_d + i_b) \cdot 2e \cdot \Delta F \cdot G^2 + \frac{4k_B \cdot \Delta f \cdot T_k}{R_L}} \tag{16}$$

where n is the average number of pulse samples, i_s is the current intensity of each channel, i_d is the dark current of the photodetector, i_b is the noise current generated by the sky stray light in the detection channel, e is the electronic charge, ΔF is the bandwidth of the optoelectronic measurement circuit, G is the gain of the photodetector, R_L is the load resistance of the photodetector measurement circuit, k_B is the Boltzmann's constant, T_k is the operating temperature of the photodetector.

V. LiDAR Preliminary Measurement

To minimize the influences of atmospheric turbulence on the detection accuracy, the whole detection for the combined BWRSL and LSRSL system is completed within 16 min. For the LSRSL system with the height resolution of 36 m, the step number is set to be 39 to realize the detection range from the ground up to the height of 1.37 km, corresponding to the maximum elevation angle of 87.1° with the angle precision of 0.1°. To increase the signal-to-noise ratio, the lateral scattering

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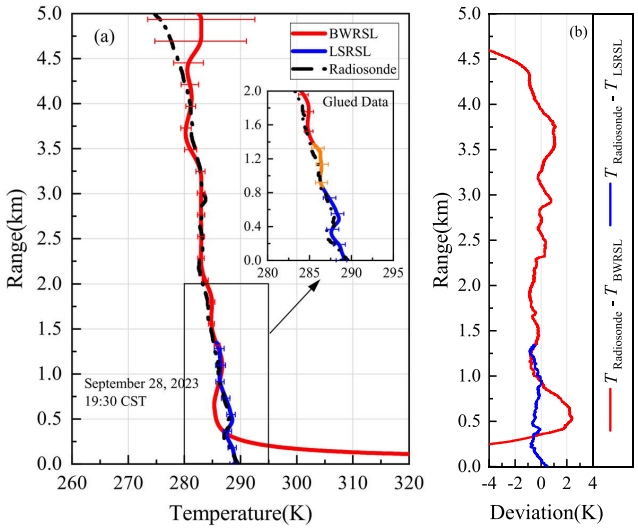


Fig. 8. Retrieval results and their deviation of atmospheric temperature using the combined BWRSL and LSRSL system. (a) Retrieval results of atmospheric temperature using the combined BWRSL and LSRSL system. The radiosonde data are plotted together for comparison. The error bars show the statistical error of temperature due to the signal noise. (b) Deviation between LiDAR system and radiosonde.

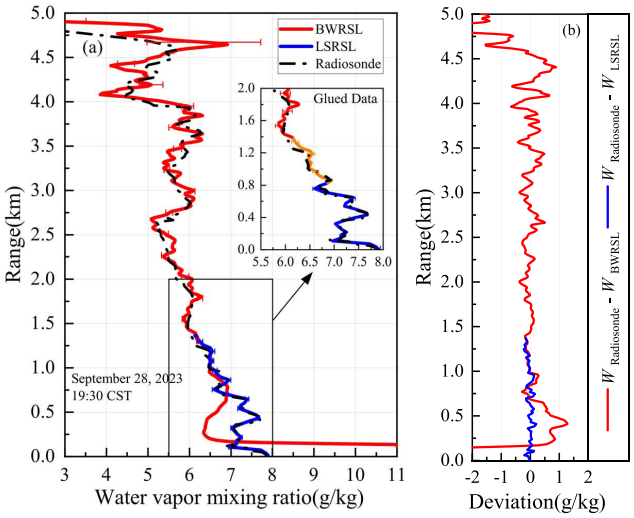


Fig. 9. Retrieval results and their deviation of water vapor mixing ratio using the combined BWRSL and LSRSL system. (a) Retrieval results of water vapor mixing ratio using the combined BWRSL and LSRSL system. The radiosonde data are plotted together for comparison. The error bars show the statistical error of water vapor mixing ratio due to the signal noise. (b) Deviation between LiDAR system and radiosonde.

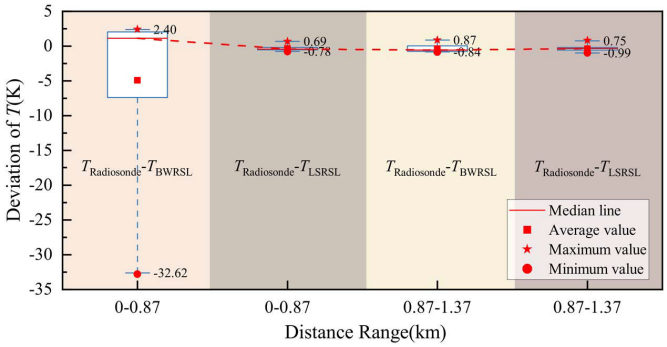


Fig. 10. Box plots of the deviations of atmospheric temperature measured by the combined BWRSL and LSRSL system in the height intervals of 0–0.87 and 0.87–1.37 km.

signal at each step of the LSRSL system is a summation result of 20 s with the synchronous triggers of the pulsed laser, while the Raman backscattering signals in all four channels of the BWRSL is a summation result of 9600 laser pulses to obtain LiDAR returns from higher heights. The noise elimination in data processing for both LiDAR systems is performed using the noise data acquisition without laser transmitting, which includes the electrical noises from PMT and A/D converter and the external environmental noise. Considering the sky stray light for the LSRSL system at different elevation angle maybe different, the noise elimination in the LSRSL system is performed using the corresponding noise data at the same elevation angle. For the BWRSL system, the noise elimination is using the collection of the corresponding noise data for each channel before each experiment.

Fig. 8 shows the retrieval results and their deviation of atmospheric temperature using the combined BWRSL and LSRSL system. The radiosonde data with the range resolution of 3 m is plotted together for comparison. The radiosonde was launched at 19:15 Chinses Standard Time (CST) on September 28, 2023 in Xi'an Atmospheric Exploration Center, which has 18 km horizontal distance with lidar site. There is a significant temperature difference between the BWRSL system and the radiosonde below the height of 0.87 km, due to the overlap effect of the BWRSL system. It is concluded that the LSRSL system can realize the atmospheric temperature detection at the heights of 0–1.37 km, while the BWRSL system can realize the detection at the heights of 0.87–4.00 km with the temperature statistical error less than 1 K. Moreover, a temperature inversion layer at the height of 0.50 km is observed by the LSRSL system, while it cannot be seen by the BWRSL system due to the height within its no-overlap regions.

Fig. 9 shows the retrieval results and their deviation of water vapor mixing ratio using the combined BWRSL and LSRSL system. The radiosonde data with the range resolution of 3 m is plotted together for comparison as well. Similar with the profile of atmospheric temperature, there is a large water vapor mixing ratio difference between the BWRSL system and the radiosonde below the height of 0.87 km, due to the overlap effect of the BWRSL system. It is concluded that the LSRSL system can realize the water vapor detection at the heights of 0–1.37 km, while the BWRSL system can realize the detection at the heights of 0.87–4.00 km with the water vapor statistical error less than 0.5 g/kg. Moreover, a water vapor accumulation layer at the height of 0.50 km is observed by the LSRSL system, while it cannot be seen by the BWRSL system due to the height within its no-overlap regions.

To better illustrate the limitations of the BWRSL system and the benefits of the LSRSL system in the lower atmosphere, Figs. 10 and 11 present the deviations of atmospheric temperature and water vapor mixing ratio measured by the combined BWRSL and LSRSL system at the height intervals

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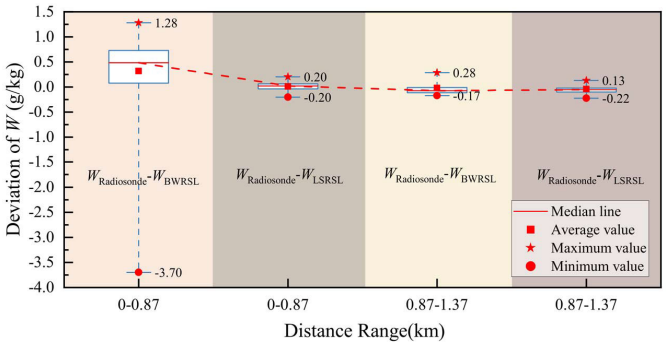


Fig. 11. Box plots of the deviations of water vapor mixing ratio measured by the combined BWRSL and LSRSL system in the height intervals of 0–0.87 and 0.87–1.37 km.

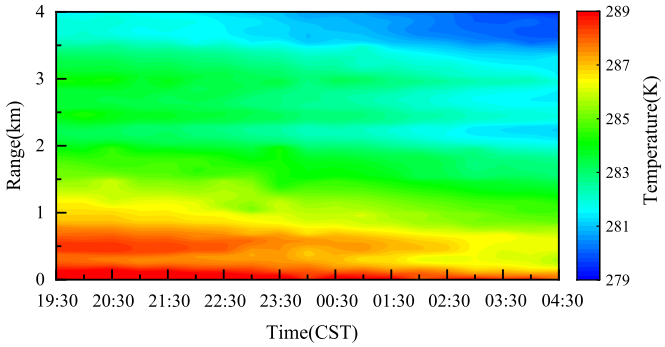


Fig. 12. THI diagram of atmospheric temperature using the glued data from the combined BWRSL and LSRSL system. The measurements are performed from 19:30 CST on September 28 to 04:30 CST on September 29, 2023.

of 0–0.87 and 0.87–1.37 km, respectively, which is represented using the colored gradient stripes.

Fig. 10 shows the deviations of atmospheric temperature for the combined BWRSL and LSRSL system in the height intervals of 0–0.87 and 0.87–1.37 km. In the height interval of 0–0.87 km, the temperature deviation of the BWRSL system is varied between −32.62 and 2.4 K due to the effect of overlap function, while the temperature deviation of the LSRSL system is fluctuated between −0.78 and 0.69 K. It shows the advantages of the LSRSL system in the detection of atmospheric temperature in the near-ground atmosphere compared to the BWRSL system. In the height interval of 0.87–1.37 km, the temperature deviation of the BWRSL system is between –0.84 and 0.87 K, while the temperature deviation of the LSRSL system is between −0.99 and 0.75 K. The very close temperature deviation between the BWRSL system and the LSRSL system presents the feasibility of data gluing to express the normalized atmospheric temperature from the ground to the concerned height.

Fig. 11 shows the deviations of water vapor mixing ratio for the combined BWRSL and LSRSL system in the height intervals of 0–0.87 and 0.87–1.37 km. In the height interval of 0–0.87 km, the deviation of water vapor mixing ratio of the BWRSL system is varied between −3.70 and 1.28 g/kg due to the influence of overlap function, while the deviation of water vapor mixing ratio of the LSRSL system is only between −0.20 and 0.20 g/kg. It shows the feasibility of the LSRSL system in the detection of water vapor mixing ratio in the near-ground atmosphere compared to the BWRSL system. In the height interval of 0.87–1.37 km, the deviation of water vapor mixing ratio of the BWRSL system is between −0.17 and 0.28 g/kg, while the deviation of water vapor mixing ratio of the LSRSL system is varied between −0.22 and 0.13 g/kg. The very close deviation of water vapor mixing ratio between the BWRSL system and the LSRSL system shows the feasibility of data gluing to express the normalized atmospheric water vapor from the ground to the concerned height.

Based on the analysis of the deviations of atmospheric temperature and water vapor, it is possible to make a data gluing between the BWRSL system and LSRSL system to minimize the retrieval deviation and express atmospheric information uniformly. The algorithm for data gluing between the BWRSL system and the LSRSL system can be expressed as

$$G(T_z, W_z, z)\begin{cases} G_{\mathrm{LSRSL}}(T_z, W_z, z), & z \le h_1 \\ \dfrac{z-h_1}{2} G_{\mathrm{LSRSL}}(T_z, W_z, z) & \\ \quad + \dfrac{h_2-z}{2} G_{\mathrm{BWRSL}}(T_z, W_z, z), & h_1 < z < h_2 \\ G_{\mathrm{BWRSL}}(T_z, W_z, z), & z \ge h_2 \end{cases} \tag{17}$$

where $G_{\mathrm{LSRSL}}(T_z, W_z, z)$ and $G_{\mathrm{BWRSL}}(T_z, W_z, z)$ represents the temperature and water vapor mixing ratio obtained by the LSRSL system and BWRSL system, respectively. The important issue for data gluing is to determine the height interval between h_1 and h_2, where the temperature deviation is less than ±1 K and water vapor deviation is less than ±0.5 g/kg. In our case, h_1 takes the height of 0.87 km, while h_2 takes the height of 1.37 km.

Fig. 12 shows the time-height-indicator (THI) plot of atmospheric temperature filled with the glued data simultaneously measured by the combined BWRSL and LSRSL system from 19:30 CST on September 28 to 04:30 CST on September 29, 2023, which are 34 profiles with a temporal resolution of 16 min and a vertical range resolution of 36 m. It can be seen that atmospheric temperature below the height of 4.00 km shows a decreasing trend with height, but the change rate of temperature along the height is indeed only 2.5 K/km. The stratified structure of the stable nocturnal boundary layer can also be seen very clearly, and there is also an inversion layer of temperature in the height range of 0.50–0.80 km from the beginning of measurements and vanished almost at 03:00 CST on September 29.

Fig. 13 shows the THI plot of atmospheric water vapor filled with the glued data simultaneously measured using the combined BWRSL and LSRSL system from 19:30 CST on September 28 to 04:30 CST on September 29, 2023, which are 34 profiles with a temporal resolution of 16 min and a vertical resolution of 36 m. It can be seen that most of water vapor exists below the height of 1.00 km, and the inconstant of water vapor mixing ratio is changed obviously before and after 0 o'clock. The water vapor distribution is characterized

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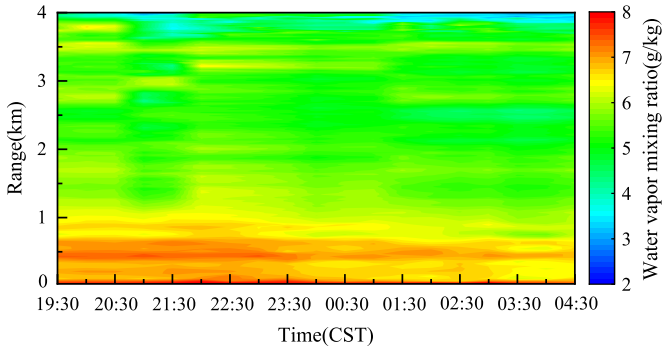


Fig. 13. THI diagram of water vapor using the glued data from the combined BWRSL and LSRSL system. The measurements are performed from 19:30 CST on September 28 to 04:30 CST on September 29, 2023.

by a clear stratified evolution, with two obvious water vapor accumulation layers at 0.50 and 0.60 km, which was lasting throughout the whole night.

VI. Conclusion

A novel Raman LiDAR system using both BWRSL technique and LSRSL technique is proposed, which can realize the accurate measurement of atmospheric temperature and water vapor from the ground to the concerned height. The most advantages of the combined BWRSL and LSRSL system are the sharing of the transmitted laser and the measurement without the effect of the geometrical overlap function. The profiling of the BWRSL system is realized by collecting the Raman backscattering signal in its dependence on the time elapsed after the transmission of the laser pulse, while the profiling of the LSRSL system is realized by the elevation angle scanning of the lateral receiver. Therefore, the BWRSL system can realize the atmospheric detection at higher heights, and the LSRSL system can realize the atmospheric detection at lower heights, whose combination results in the detection height of atmospheric temperature and water vapor from the ground to the concerned height. The continuous measurements using the combined BWRSL and LSRSL system can be used for investigating the atmospheric structure revolution of the nocturnal stable boundary layer, studying the pollution transport within and out of atmospheric boundary layer, as well as the prediction of the frost to protect the agricultural cash crops, and so on.

In the polychromator design of both BWRSL system and LSRSL system, the optical cage-type light-blocking cylinder is designed to be the detection module with the collimating lens, narrow-band interference filter and focusing lens, which is equipped in the detection channel to separate and extract the desired Raman scattering signals for both LSRSL system and BWRSL system. The consistency design of the detection module brings great convenience to the integration of the combined BWRSL and LSRSL system. Moreover, to increase the detection accuracy, the optimal center wavelengths of the high- and low-quantum-number transitions of PRRSS and VRSS of N_2 and H_2O are discussed as well, which are 533.24 nm in the low-quantum-number rotational Raman channel, 535.52 nm in the high-quantum-number rotational Raman channel, 607.44 nm in the detection channel of the VRSS of N_2, and 660.43 nm in the detection channel of the VRSS of H_2O.

After system construction, the preliminary experiments are carried out with the temporal resolution of 16 min and the vertical resolution of 36 m. The measurement results show that the LSRSL system can realize the detection heights of 0.00–1.37 km and the BWRSL system can realize the detection heights of 0.87–4.00 km, respectively, with the temperature deviation and statistical error less than 1 K and water vapor mixing ratio deviation and statistics error less than 0.5 g/kg. The continuous observations of atmospheric temperature and water vapor filled with the glued data show the layered structure and the detailed variation of the stable nocturnal boundary layer.

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